Notes: Iwatsuki (1992) — *Stentor coeruleus* Shows Positive Photokinesis

Source: Iwatsuki, K. (1992). *Stentor coeruleus* shows positive photokinesis. *Photochemistry and Photobiology*, 55(3), 469–471. (PDF: `references/Iwatsuki-1992-Stentor-coeruleus-positive-photokinesis.pdf`)

Key findings

- *S. coeruleus* shows **positive photokinesis**: swims faster in light, slower in dark. Mean velocity ~0.6 mm/s at 100 lx, up to ~1.0 mm/s at 50000 lx (white light).
- Velocity increases roughly **linearly with log(fluence rate)** — a Weber–Fechner-type relationship, not linear in raw light intensity.
- Photokinesis has its own action spectrum, distinct from the photophobic-response/ phototaxis action spectrum (peak 610 nm). This implies **at least two separate photoreceptor systems** in the cell.
- *S. coeruleus* is reported as the first protozoan shown to exhibit all three photoresponse types: photophobic response, phototaxis, and photokinesis.

Figure 3

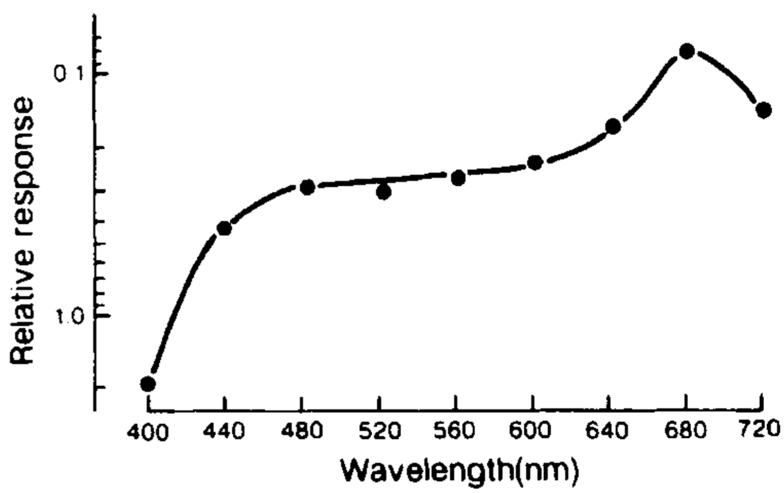

Figure 3. Action spectrum for photokinesis in *S. coeruleus*. The fluence rate of monochromatic light at nine different wavelengths which gave the half-maximal kinetic response (0.58 mm/s) was determined to obtain the action spectrum.

How Figure 3 (action spectrum) was built

For each of 9 wavelengths (400–720 nm), the authors measured a full fluence-rate vs. velocity dose-response curve (like the 680 nm / 440 nm examples in Fig. 2a — velocity rises ~linearly with log fluence rate). From each curve they read off the **fluence rate required to reach a fixed criterion velocity: 0.58 mm/s** (the paper calls this the "half-maximal kinetic response," but see the "Why 0.58 mm/s" section below — the paper does not describe how this specific number was derived, and it's likely just a common target velocity within the overlapping range of all 9 curves rather than a computed dark-floor/light-ceiling midpoint). This lets 9 whole dose-response curves collapse into 9 single comparable numbers (essentially an EC50-per-wavelength), which is what's plotted as the action spectrum.

Reading the y-axis: logarithmic AND inverted

- Y-axis label: "Relative response." Y-axis units are actually the **fluence rate (W/m^2)** needed to hit 0.58 mm/s — same scale as the x-axis of Fig. 2a (log, ~0.1–1.0+ W/m^2).
- **Where "W/m^2" comes from:** Fig. 3's own axis and caption never print a unit — the caption only says "the fluence rate of monochromatic light… was determined." The W/m^2 unit is confirmed indirectly, from Fig. 2's intensity-axis labels ("0.1 W/m^2 (400 nm)", "0.5 W/m^2 (400 nm)", "1.0 W/m^2 (400 nm)" per the OCR'd PDF text), which measure the same quantity (fluence rate of monochromatic light) over the same ~0.1-1.0+ numeric range as Fig. 3's y-axis. So the unit is inferred by matching quantities across figures, not read directly off Fig. 3.
- The axis is drawn **inverted** (small numbers near the top, large numbers near the bottom). This is a standard action-spectrum convention: the biologically meaningful quantity is *sensitivity* = 1/fluence needed, not the fluence itself. Rather than plot the reciprocal, the axis is just flipped so "higher on the page" = "less light was needed" = "more sensitive" — producing a normal-looking peaked curve even though the raw printed numbers are fluence rates, not sensitivity units.

Specific values discussed

- **680 nm to ~0.08 W/m^2** needed to reach 0.58 mm/s (near the top of the plot — most sensitive point tested).
- **400 nm to ~1.1 W/m^2** needed to reach 0.58 mm/s (near the bottom — least sensitive point tested).
- Ratio ~= **14x** — Stentor needs roughly 14 times less red (680 nm) light than blue (400 nm) light to produce the same swimming-speed increase. This is a genuine, strong action-spectrum peak at 680 nm, not just a mild bias.
- Curve shape across the spectrum: steep rise from 400–440 nm, a plateau through ~440–600 nm, then a second climb to the true peak at 680 nm,

followed by a drop at 720 nm.

Why 0.58 mm/s specifically

Correction / uncertainty flag: initial notes here speculated that 0.58 mm/s was the midpoint between a measured "dark floor" velocity and a measured "saturating ceiling" velocity. That is *not* stated anywhere in the paper's Materials & Methods and is likely wrong, for two reasons:

1. **A true zero-light data point isn't measurable with their setup.** The same light source that provides the photokinetic stimulus also provides the dark-field illumination the "Bug-tracker" camera needs to see/track the cell at all ("To generate dark-field illumination and to stimulate the organisms, white light… was… projected…"). There's no separate viewing-light channel, so they can't record a cell swimming in literal darkness.
2. **No plateau is described.** Both Fig. 1a (white light, 100–50000 lx) and Fig. 2a (monochromatic, ~0.1–2 W/m^2) are described as "increased almost linearly with a logarithmic increase" across the *entire* tested range — no floor/ceiling saturation is mentioned, so there's no natural min/max to average.

More likely explanation: 0.58 mm/s is a fixed criterion velocity chosen because it falls *within the overlapping, actually-measured range* of all nine wavelength-specific fluence-response curves — i.e., a common target every curve crosses somewhere in its tested data, so a fluence-rate value can be read off (interpolated, not extrapolated) for each of the 9 wavelengths. This is the standard practical reason for picking one criterion response when collapsing several dose-response curves into a single action spectrum; it does not require or imply any dark/saturating-max calibration measurement.